

Project Prioritization (MCDA)

Generated by uber-polya

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Multi-Objective Optimization

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

Rank 8 software features by priority using 4 weighted criteria: user impact (weight 0.4), development effort (0.25, lower is better), strategic alignment (0.2), and revenue potential (0.15). Each feature is scored 1-10 on each criterion. Produce a defensible priority ranking.

Real-World Mapping

Real-World Concept	Mathematical Object
total score/cost	objective function value

Constraints

- | | | |
|-----|-------------------------|--|
| (1) | MCDA / weighted scoring | structured approach to multi-objective decision making |
| (2) | Score normalization | making heterogeneous criteria comparable |
| (3) | Criterion inversion | handling "lower is better" metrics |
| (4) | Sensitivity analysis | testing robustness of rankings to weight changes |
| (5) | Transparency | full breakdown of how each feature scored |

Objective

$$\max \sum_{i,j} s_{ij} x_{ij}$$

Approach

Suggested approach MCDA weighted linear scoring with min-max normalization

Complexity class:

Available tools: MCDA weighted linear scoring with min-max normalization

2. Solution

Answer:	Solution computed
Objective Value:	0.8229
Optimal:	Yes
Feasible:	Yes
Algorithm:	MCDA weighted linear scoring with min-max normalization
Time:	0.0003s
Certificate:	Solver reports optimal.

Solution Details

Objective value: $z^* = 0.8229$

Method: Multi-Criteria Decision Analysis (MCDA) with weighted linear scoring.

- Normalize raw scores to [0, 1] range using min-max normalization
- Invert effort scores (lower effort is better)
- Apply criterion weights and compute weighted composite score
- Rank features by composite score

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	MCDA weighted linear scoring with min-max normalization

3. Interpretation

The Question

Rank 8 software features by priority using 4 weighted criteria: user impact (weight 0.

The Answer

The optimal solution has objective value $z^* = 0.8229$.

What This Means

- Ranked list of 8 features with composite scores
- Normalized score breakdown per criterion
- Sensitivity analysis: how rankings change if weights shift
- Independent verification of score computation

Multi-Criteria Decision Analysis (MCDA) with weighted linear scoring.

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Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution